ModelIt!

Teacher's Guide

Why Is Lithium the New Gold?

Modeling the Resource Economics of the Electric Vehicle Revolution

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-ESS3-1, HS-PS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-1, HS-PS1-3

Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity. Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the structure of atoms and molecules.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of lithium supply dynamics and design resource management strategies informed by computational model data.
Disciplinary Core Idea	**ESS3.A Natural Resources / PS1.A Structure and Properties of Matter** — The availability of lithium as a natural resource is constrained by geological distribution, extraction technology, and the atomic-level properties that make it uniquely suited for batteries.
Crosscutting Concept	**Cause and Effect / Systems and System Models** — Students trace causal chains from EV policy decisions through resource extraction to environmental consequences, modeling the lithium economy as an interconnected system with feedback loops.

CAST Testing Alignment

MULTIPLE CHOICE:

Global EV sales triple over five years while lithium mining capacity increases by only 40%. Based on your model, which outcome is MOST likely?

CONSTRUCTED RESPONSE:

Using your model, explain why simply opening more lithium mines is not a sustainable long-term solution to the supply crisis. Reference at least three model components and their interactions in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Lithium — element number 3, the lightest metal on the periodic table — has become the most strategically important mineral of the 21st century. Every electric vehicle battery, grid-scale energy storage system, and smartphone depends on it. As the world races to decarbonize transportation, the demand for lithium is skyrocketing while supply chains buckle under the pressure.

Why Lithium? The Atomic Advantage

Lithium's dominance in batteries comes from its atomic properties: it has the highest electrochemical potential of any element, the lowest atomic mass of any metal, and a small ionic radius that allows lithium ions to move quickly through electrode materials. These properties produce batteries with unmatched energy density — storing more energy per kilogram than any competing chemistry. A single EV battery contains 8-12 kg of lithium, and global EV sales are projected to reach 40 million per year by 2030.

The Extraction Problem

About 60% of the world's lithium comes from brine deposits in the 'Lithium Triangle' of Chile, Argentina, and Bolivia. Brine extraction pumps underground saltwater into massive evaporation ponds where the sun concentrates lithium over 12-18 months. This process consumes approximately 2.2 million liters of water per ton of lithium — in some of the driest places on Earth. In Chile's Atacama Desert, lithium mining consumes 65% of the region's water, devastating indigenous farming communities and fragile ecosystems. Hard-rock mining in Australia is faster but generates significant toxic waste.

The Price Rollercoaster

Lithium markets are extraordinarily volatile because supply is inelastic — new mines take 7-10 years from discovery to production. When Tesla and other manufacturers announced massive EV expansion plans, lithium carbonate prices surged from $6,000/ton in early 2021 to $80,000/ton in late 2022. When demand growth briefly paused, prices crashed back to $12,000/ton by 2024. This volatility makes long-term planning nearly impossible for both miners and manufacturers.

The Recycling Imperative

Currently, less than 5% of lithium-ion batteries are recycled. Most end up in landfills, leaching toxic materials into soil and groundwater. But recycled lithium requires 90% less water and produces 75% less CO_2 than newly mined lithium. Companies like Redwood Materials and Li-Cycle are building facilities that can recover 95% of lithium from spent batteries. The challenge is building recycling capacity fast enough — the first massive wave of EV batteries will reach end-of-life around 2030-2035.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the lithium economy system: Lithium Demand, Mining Rate, Environmental Damage, Battery Performance, Recycling Rate, Market Price, and EV Adoption — distinguishing external drivers from internal system responses.

E **Establish Relationships**

Students determine that EV Adoption drives Lithium Demand, which outpaces Mining Rate, causing Market Price spikes. Mining Rate increases Environmental Damage. High Market Price incentivizes Recycling Rate investment. Better Battery Performance reduces Lithium Demand per vehicle.

V **Visualize & Model**

Students build a computational model showing the interconnected feedback loops of the lithium supply chain, visualizing how shocks propagate through the system.

E **Evaluate Outcomes**

Students run steady growth, EV boom without recycling, and circular economy scenarios to identify which strategies resolve the supply-demand-environment trilemma.

R **Revise & Extend**

Students add Geopolitical Risk, Alternative Chemistry, or Water Scarcity Index to explore more realistic supply chain dynamics and vulnerability analysis.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This metal costs more than some people's rent per gram. It powers every EV, every phone, every laptop. And we're running out of easy ways to get it."*

Watch For

Show lithium price chart from 2020-2025. Let the volatility speak for itself. Ask: What could cause a price to swing this wildly?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling one of the most important supply chain crises of your generation — the race for lithium."*

Watch For

Have students read objectives. Pre-teach 'critical mineral' and 'circular economy.' Quick-write: Name three things that use lithium batteries.

Slide 3: Big Question

3 min

Say: *"Gold is valuable because it's rare and beautiful. Lithium is valuable because without it, the entire green energy transition stops. Which matters more?"*

Watch For

Poll: How many devices with lithium batteries are in this room right now? Count them. Multiply by the lithium content.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of an entire global supply chain — from mines in Chile to car dealerships in America."*

Watch For

Preview each LEVER step. Emphasize that this is a system with feedback loops, not a simple supply-demand line.

Slide 5: Activity 1: Components

8 min

Say: *"Seven moving parts in this system. Some you can control with policy, some respond to the market. Which is which?"*

Watch For

Guide sorting of external vs. internal components. Discuss why Recycling Rate and EV Adoption are external (policy-driven) while others respond to system dynamics.

Slide 6: Activity 2: Connections

8 min

Say: *"When EV sales double, what chain reaction ripples through the entire lithium economy? Trace it."*

Watch For

Help students map feedback loops. Identify the key tension: EV Adoption drives demand but Mining Rate can't keep up. Find the stabilizing role of Recycling Rate.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's crash-test the lithium economy. What happens if everyone buys an EV but nobody recycles the batteries?"*

Watch For

Students predict outcomes first, then run all three scenarios. Compare the circular economy scenario to the boom-without-recycling scenario.

Slide 8: Discoveries

5 min

Say: *"The green energy transition has a dirty secret — and your model just exposed it. But it also found the solution."*

Watch For

Lead discussion on the paradox of mining for sustainability. Connect to real-world lithium recycling companies and policy proposals.

Slide 9: STEM Challenge

12 min

Say: *"An EV company just hired your team. You have 20 years and a billion-dollar budget. Design a lithium strategy that doesn't destroy the planet."*

Watch For

Groups design comprehensive supply strategies balancing cost, environment, and security. Present to class as board proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Recycling Rate, EV Adoption
- **INTERNAL:** Lithium Demand, Mining Rate, Environmental Damage, Battery Performance, Market Price

Relationships

- **EV Adoption to Lithium Demand:** POSITIVE (+) — Higher EV Adoption directly increases Lithium Demand because each electric vehicle requires 8-12 kg of lithium for its battery pack. As more consumers switch from gasoline to electric, total lithium consumption scales proportionally.
- **Lithium Demand to Market Price:** POSITIVE (+) — When Lithium Demand exceeds Mining Rate capacity, Market Price rises sharply due to competition for scarce supply. Because lithium supply is inelastic in the short term, even modest demand increases can cause dramatic price spikes.
- **Mining Rate to Environmental Damage:** POSITIVE (+) — Higher Mining Rate directly increases Environmental Damage through greater water consumption in brine extraction, more toxic waste from hard-rock processing, expanded habitat destruction, and increased displacement of indigenous communities near mining sites.

Key Discoveries

- Lithium supply cannot keep pace with projected EV demand because new mines take 7-10 years to develop while demand is growing 25-35% annually
- The environmental cost of lithium extraction creates a painful paradox — the green energy transition requires mining practices that damage ecosystems and water supplies
- High Market Price triggers both increased mining investment AND increased recycling investment, but on very different timescales — recycling infrastructure can scale faster
- A circular economy approach combining improved Battery Performance (less lithium per vehicle) and high Recycling Rate is the only path that resolves the supply-demand-environment trilemma

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Electric vehicles are completely green"	While EVs produce zero tailpipe emissions, their environmental impact depends heavily on how battery materials are sourced. Lithium mining consumes millions of liters of water in arid regions, generates toxic waste, and displaces communities. The full life cycle analysis shows EVs are still significantly cleaner than gasoline cars, but they are not zero-impact.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We'll run out of lithium soon"	Global lithium reserves are actually substantial — an estimated 98 million tons. The problem isn't total supply but extraction rate. Current mining capacity cannot scale fast enough to meet projected demand. It's a flow problem, not a stock problem. New deposits are still being discovered, and ocean water contains 230 billion tons of lithium — the challenge is extracting it economically.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Recycling batteries is too expensive to be practical"	Early lithium-ion recycling was indeed uneconomical when lithium was cheap. But with prices now 5-10x higher than a decade ago and improving recycling technology, recovered lithium is increasingly cost-competitive with mined lithium. Recycling also avoids the 7-10 year development timeline of new mines and produces 75% less CO_2 than primary extraction.